Linking MIDAS Data with Biological Databases:

An R-Based Pipeline for Protein-Metabolite Discovery

Kainoa Martinez[1,2], McKenna Wilson[1], Paul Stewart[1], Kevin Hicks[1*]*
*[1]University of Utah, University College London[2], Corresponding Author**

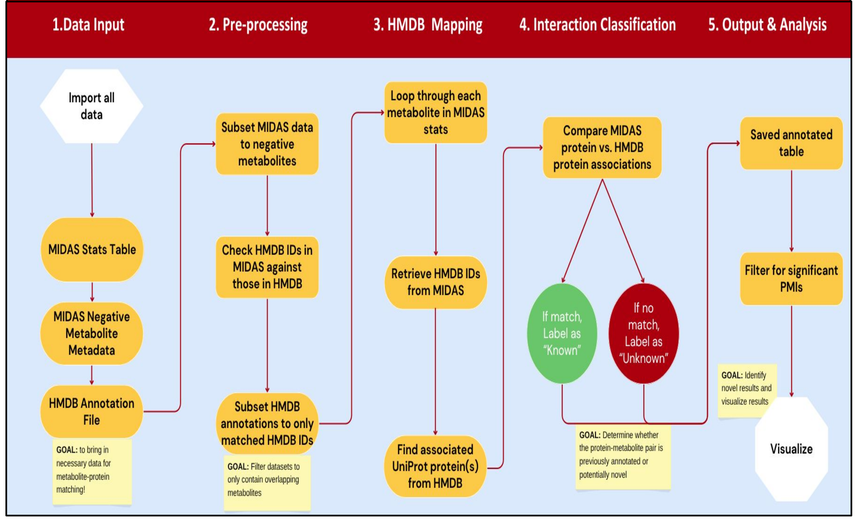
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I. Introduction

- Mapping protein-metabolite interactions (PMIs) is essential for understanding metabolism, but traditional omics approaches (e.g., proteomics, metabolomics) alone offer an incomplete picture
- MIDAS (Mass spectrometry Integrated with equilibrium Dialysis for the discovery of Allostery Systematically) enables systematic identification of protein-metabolite binding events, providing richer interaction profiles
- **Integrating MIDAS data with UniProt (Universal Protein Resource) and HMDB (Human Metabolome Database) annotations enables the discovery of known and novel protein-metabolite interactions (PMIs), revealing domain-specific metabolite-binding patterns that may underlie disease-related functions**

II. Methods



III. Results

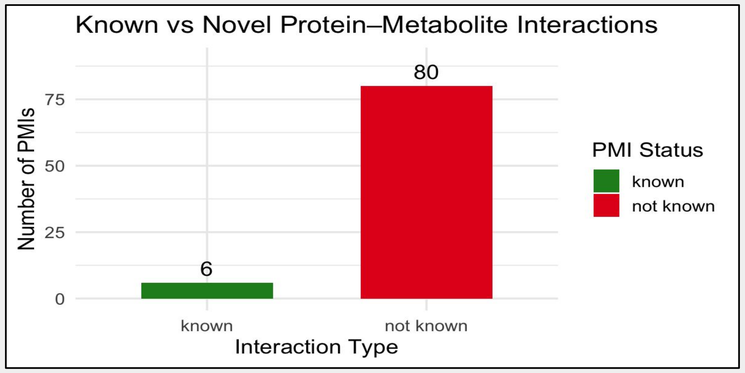


Figure 1: Distribution of Known vs. Novel Protein–Metabolite Interactions (PMIs).
Bar plot showing the number of identified PMIs classified as either previously known (green) or novel (red). Out of 86 total interactions, the majority (80) were not previously annotated in existing biological databases, highlighting the potential for novel discovery in protein-metabolite association mapping.

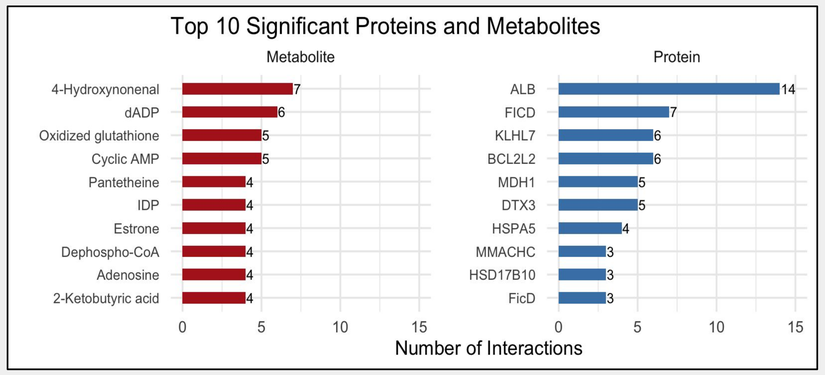
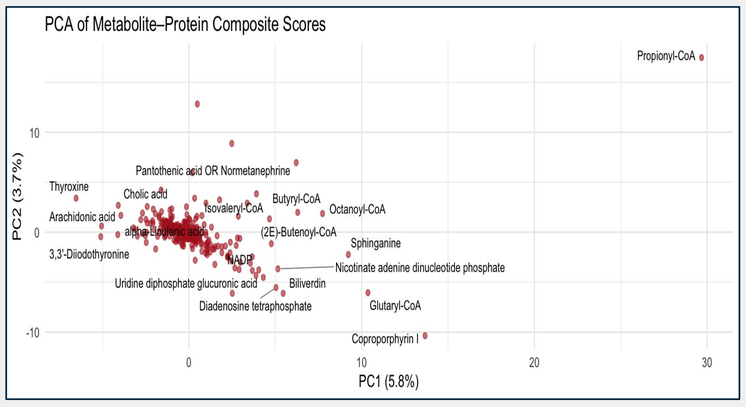


Figure 2: Known vs. Novel Protein–Metabolite Interactions (PMIs).
Bar plot showing the top 10 proteins and metabolites with the highest number of significant interactions ($log_2FC > 1$, $q < 0.01$), revealing potential hubs in the PMI network and candidates for further functional analysis.



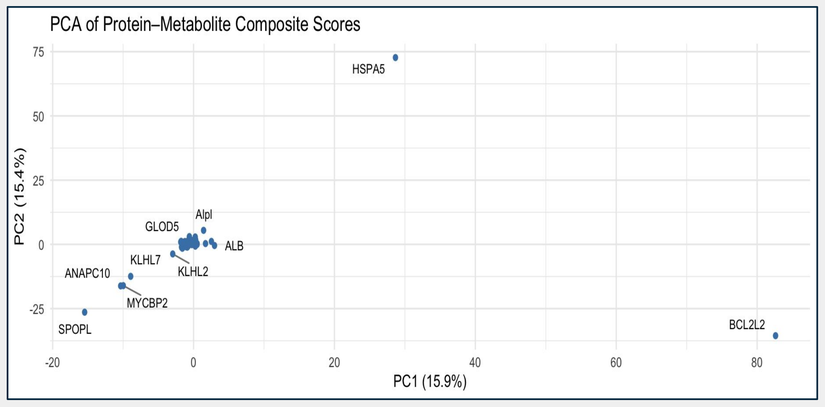


Figure 3: PCA of Protein–Metabolite Composite Scores
Principal component analysis (PCA) of composite interaction scores (log_2 fold change × $–log_{10}$ p-value) captures variation in protein–metabolite interaction profiles. In the **left PCA**, each point represents a **metabolite**, and in the **right PCA**, each point represents a **protein**. Distance from the origin reflects interaction distinctiveness. **Propionyl-CoA** emerges as a metabolite outlier, suggesting broad or strong associations, while **HSPA5** is a protein outlier, potentially indicating unique or selective binding activity.

IV. Conclusion

- Integrating MIDAS data with HMDB and UniProt annotations reveals a large number of novel PMIs
- Most significant PMIs are previously unreported, highlighting the value of this approach in expanding functional annotations
- PCA shows distinct clustering of interaction profiles, suggesting biological relevance
- The projects future direction aims to incorporate databases like BRENDA and STITCH to enhance the depth of protein-metabolite annotations and allow the framework to handle larger and more diverse datasets

V. Acknowledgments

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VI. Sources

Hicks, K. G., et al. (2023). Protein-metabolite interactomics of carbohydrate metabolism reveal regulation of lactate dehydrogenase. *Science, 379*(6636), 996–1003. https://doi.org/10.1126/science.abm3452

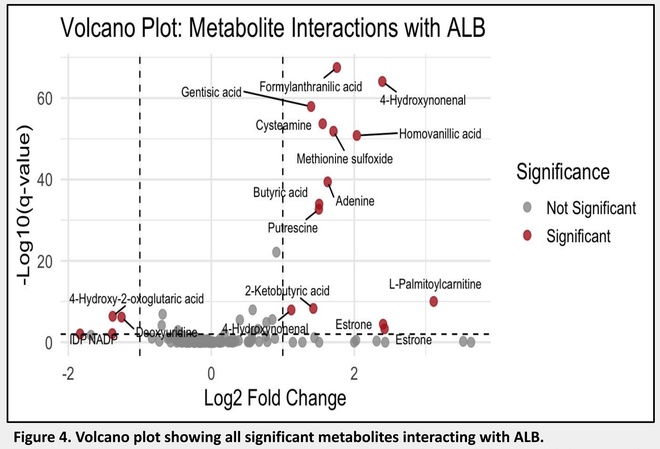


Figure 4. Volcano plot showing all significant metabolites interacting with ALB.
Each point represents a metabolite, with the x-axis indicating log2 fold change and the y-axis showing –log10(q-value). Metabolites meeting significance thresholds (|log2FC| > 1 and q-value < 0.01) are highlighted in red and labeled. Dashed lines denote the fold-change and q-value cutoffs. Among these, **4-Hydroxynonenal** appears as the most frequently interacting metabolite with ALB.

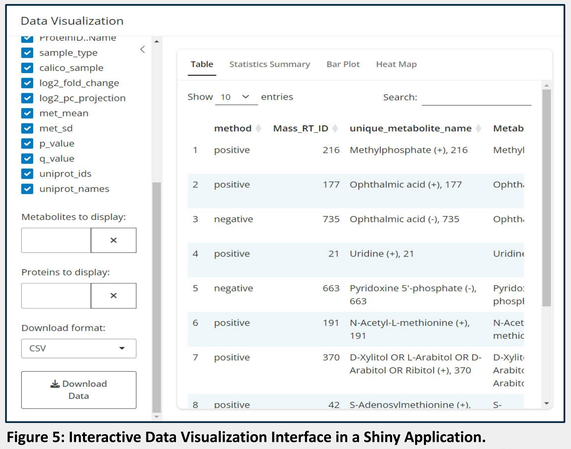


Figure 5: Interactive Data Visualization Interface in a Shiny Application.
Exploration of metabolomics and proteomics data. The left panel includes filtering options such as variable selection, metabolite/protein search, and data download tools. The main panel displays a searchable, sortable table of metabolite data, Additional tabs provide access to summary statistics, bar plots, and heat maps for further analysis.